

ECE 657D: Research Project

Faithfulness-Guided Reranking for Abstractive Summarization

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Q1: Research Question

Can simple faithfulness-based reranking strategies improve the consistency of pre-trained abstractive summarization models without significantly degrading summary quality?

To answer this question, I will generate multiple candidate summaries using a pretrained BART model and compare different reranking strategies based on faithfulness metrics. I will evaluate whether reranking improves faithfulness while maintaining competitive ROUGE scores.

Q2: Prior Work and Limitations

Abstractive summarization models often generate fluent summaries that contain factual inconsistencies or hallucinated information. Prior work has shown that standard metrics such as ROUGE are not sensitive to these errors [Maynez et al., 2020]. This limits the reliability of pretrained summarization systems in real-world applications.

Pretrained encoder-decoder models such as BART [Lewis et al., 2020] achieve strong performance on benchmarks like CNN/DailyMail [See et al., 2017] and XSum [Narayan et al., 2018]. However, these models frequently produce summaries that are not fully faithful to the source document [Maynez et al., 2020].

Several automatic faithfulness metrics have been proposed. Wang et al. [2020] introduced QAGS, a question-answering based metric, which correlates better with human judgment but is computationally expensive and can be brittle if the QA model itself fails. Kryscinski et al. [2020] developed FactCC to check consistency using synthetic data, yet it often fails to generalize to the subtle paraphrasing found in real-world news summaries. Laban et al. [2022] proposed SummaC, which improves on NLI-based consistency checking, but it still struggles with "granularity" where a single hallucinated word might be missed by sentence-level checks.

While these metrics improve faithfulness evaluation, they are typically used for post-hoc scoring rather than selection. And there's limited systematic comparison of lightweight reranking rules under fixed-generator, small-compute settings.

Q3: Novelty of Approach

My project offers an original contribution through a systematic empirical comparison and a novel Agreement-gated reranking: selects a candidate only when at least two faithfulness metrics agree on the same best candidate in the n-best list. If no candidate satisfies the agreement gate, we fall back to selecting the candidate with the highest weighted sum of normalized faithfulness scores (e.g. equal weights, z-score)

This specific gating mechanism has not been widely benchmarked against standard beam search [Freitag and Al-Onaizan, 2017].

Q4: Impact and Motivation

Faithfulness is a major bottleneck for deploying abstractive summarization systems in high-stakes domains such as news and healthcare. Prior work shows that neural summarizers frequently hal-

lucinate unsupported facts [Maynez et al., 2020], and standard metrics like ROUGE are often insensitive to such errors [Kryscinski et al., 2020].

If simple reranking strategies can improve factual consistency without retraining large models, this would offer a lightweight and computationally inexpensive way to increase reliability. More broadly, this project helps clarify the trade-off between surface-level overlap metrics and factual consistency in summarization.

Q5: Risks and Assumptions

This project depends on faithfulness metrics to evaluate factual consistency, but these metrics are only approximations of human judgment and may not perfectly capture factual errors. It is possible that reranking improves faithfulness while slightly lowering ROUGE, leading to a clear trade-off between quality and factual consistency. Another risk is that beam search may not produce sufficiently diverse candidates; this approach assumes that the n -best list includes at least one candidate that is more faithful than the default top-1 output.

I assume that the selected faithfulness metrics capture complementary aspects of factual consistency rather than overlapping noise, so combining them is meaningful. Also, I assume that evaluating 300–500 validation examples is sufficient and representative enough to observe practically meaningful differences between reranking strategies.

Q6: Resources

Experiments will be conducted on the CNN/DailyMail dataset [See et al., 2017] and XSum [Narayan et al., 2018] separately (CNN/DailyMail summaries are relatively extractive, while XSum summaries are highly abstractive and prone to hallucination), using 300–500 validation examples for evaluation. The base model will be pretrained BART-large [Lewis et al., 2020]. All experiments are inference-only with a limited beam size (e.g., $k = 5$), making the setup computationally feasible. Compute will be provided via Kaggle Notebooks (30 hours/week of GPU access), which is sufficient for the proposed reranking experiments.

Q7: Weekly Plan

Week	Plan
1 (Mar 3–7)	Set up Kaggle; load datasets; generate BART top-1 and n -best ($k = 5$) summaries.
2 (Mar 10–14)	Implement faithfulness metrics; compute ROUGE and faithfulness for baseline.
3 (Mar 17–21)	Implement reranking: single-metric and agreement-gated selection.
4 (Mar 24–28)	Evaluate methods; compare to baseline; analyze ROUGE–faithfulness trade-offs.
5 (Mar 31–Apr 2)	Qualitative analysis; finalize figures and complete report.

Q8: Milestones and Success Criteria

The baseline is the standard BART-large top-1 beam output ($k = 5$). Performance will be evaluated using ROUGE-1, ROUGE-2, ROUGE-L, and the faithfulness metrics.

The project will be considered successful if the reranking strategy improves faithfulness relative to the baseline while keeping ROUGE-L degradation within 1–2 absolute points. Beyond raw improvements, success also requires a clear and systematic empirical comparison that characterizes the trade-off between summary quality and factual consistency.

References

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